Slop / University

Lit for Whoever's Moving: Validating a Sports Oval's Motion-Sensing Floodlight Extension Against Independent Occupancy Ground Truth

Osei Vandermeer, Thandiwe Solberg

School of Continuous Improvement, Slop University

doi:10.5555/slop.hed0q8

Abstract. When a booked evening session on the University's main outdoor sports oval runs past its scheduled end time, the floodlight controller keeps the towers lit rather than cutting to black on the whistle, extending in short increments for as long as its motion-sensor array reports movement inside the enclosure. The extension logic assumes that motion at the light towers means a player still on the ground. We test that assumption by pairing eleven weeks of the controller's own extension log against an independent record the grounds team already keeps: the on-duty supervisor's time-stamped closing-log entry, made without reference to the floodlight system, noting when the oval was confirmed clear. Across 132 extension events triggered over 96 evening bookings, only 46% coincided with a supervisor-confirmed occupant; the remainder are attributable, from a maintenance camera's timestamped stills, to grounds-maintenance passes, irrigation cycles, wildlife at the verge, and insect swarms around the lamp housings. A contractor-recommended sensitivity-threshold increase, trialled for the final four weeks, moved the agreement rate by two percentage points (n.s.). We discuss what a motion-triggered courtesy feature is actually protecting against once its own sensor loop is the only witness asked.

Introduction

A booked evening session that runs long used to mean one of two things for anyone still on Slop University's main outdoor sports oval: finish in the dark, or watch the lights cut mid-play exactly on the hour. A 2025 retrofit added a courtesy extension to the floodlight controller, so that when a booking's scheduled end time arrives the towers stay lit for as long as a tower-mounted motion-sensor array keeps reporting movement inside the enclosure, stepping down only once the ground reads clear. The feature was built, and is maintained, on a single assumption: that motion at the light towers after the whistle means a player still needs to see. Nobody has separately asked what else moves under four floodlight towers after dark, or how often the sensor's answer and the true answer part company.

That gap matters because the extension log is also the record used to justify the feature: an audit of how often the lights stayed on for a good reason, that consults only the sensor that decided the reason was good, is not an audit so much as a mirror. This paper pairs eleven weeks of the floodlight controller's own extension events against an independent signal the grounds team already produces without reference to it — the on-duty supervisor's time-stamped closing-log entry — to ask whether the extension is protecting the population it is built for.

Our contributions, each modestly hedged: (1) we report, to our knowledge, the first paired instrumentation of a campus floodlight motion-sensing extension against an independently recorded occupancy ground truth; (2) we quantify the share of extension events attributable to a cause other than a lingering player, using a maintenance camera's timestamped stills to adjudicate; and (3) we test whether a straightforward sensitivity-threshold increase — the installer's own recommended remedy for a nuisance-trigger complaint — meaningfully improves the agreement rate. It does not, and we discuss what that suggests about where the fix for a courtesy feature like this one actually lives.

Related work

Passive-infrared (PIR) motion sensing is the default technology behind occupancy-triggered lighting, prized for its low cost and its avoidance of a camera's privacy concerns [1], [2]. That convenience carries a documented cost: PIR-based occupancy systems validated against an independent ground truth routinely report substantial disagreement, driven by thermal draughts, ambient heat sources, and detection ranges that vary with target size [3], [4], [5]. Laboratory and field evaluation frameworks built specifically to characterise this gap find accuracy that depends heavily on space type, sensor placement, and the failure scenario tested, rather than settling on one trustworthy figure [6], [7]. Outdoor lighting adds its own failure modes: reviews of adaptive streetlight systems note that motion-triggered dimming schemes are tuned against vehicle and pedestrian traffic and are rarely validated against what else crosses a detection zone after dark [8]. Camera-trap ecology has faced a structurally similar problem for longer, and has built a literature around the gap between a sensor's trigger and an animal's true presence, whether the fix is a coding protocol, an embedded filter, or a purpose-built review pipeline [9], [10], [11]. Two adjacent literatures explain who else is likely on the other side of a false trigger: insects are drawn to artificial light sources in large numbers through positive phototaxis [12], and vertebrate wildlife measurably reorganises its activity around artificial light at night, including around fixed outdoor installations left running after their intended use [13], [14]. Indoor occupant-centric control work shows that pairing a sensor log against an independently collected signal, rather than trusting the sensor's own log as ground truth, is itself the more informative comparison [15] — the design this paper borrows and points outward at a floodlight tower instead of a desk. Within the University, sensor logs have separately been found to diverge from independent observation once a feedback signal enters the loop, whether a hand dryer's indicator light shaping how long hands linger despite unchanged airflow [16], or an occupancy-relevant swap that left session length and peak concurrent occupancy unchanged across a change in what was logged [17].

Method

Setting. The University's main outdoor sports oval is lit by four floodlight towers, each carrying a PIR motion-sensor array added in a 2025 controller retrofit. Under the retrofit's courtesy-extension logic, when a booking's scheduled end time is reached the controller polls each tower's sensor on a rolling window:

```
on booking_end_time:
  while any(tower.motion(window: 2min) for
tower in towers):
    extend(2min)
  step_down()
```

Detected motion within the last two minutes extends the lights by a further two minutes; two consecutive clear polls (four minutes with no detected motion at any tower) step the lights down. The logic is booking-agnostic: it fires identically whether the movement it detects belongs to a departing team or a startled fox.

Data. We draw on three independent records for every evening booking across an eleven-week teaching-period window. First, the floodlight controller's own extension log: the onset timestamp, triggering tower, and total duration of every extension. Second, the on-duty grounds supervisor's closing log — an existing paper-and-clipboard routine, unrelated to the floodlight retrofit, in which the supervisor records the clock time at which they personally confirmed the oval clear before locking the perimeter gate each night. The supervisor does not consult the floodlight log when completing it, and the floodlight controller does not consult the supervisor. Third, a tower-mounted maintenance camera installed for insurance purposes, which captures a timestamped still every five minutes; where an extension event's onset fell before the supervisor's confirmed-clear time, we used the nearest stills to code the visible cause into one of five categories — person present, grounds-maintenance activity, irrigation-cycle start, wildlife, or insect swarm at the lamp housing — with a second coder resolving the 14 cases (11%) where the first coder's call was marked uncertain.

Agreement rate.

$$\text{agreement rate} = \frac{N_{\text{confirmed}}}{N_{\text{extensions}}}$$

Sample. Of 214 evening bookings scheduled across the window, 96 ran to their advertised end time without an early finish or a same-slot overrun into the next booking, and form the analytic sample; the remainder ended before triggering the extension logic at all and carry no extension event to adjudicate. These 96 bookings produced 132 extension events (a booking could clear, re-trigger, and clear again).

Ablation. For the final four of the eleven weeks, the installer raised each tower's PIR sensitivity threshold at the on-site contractor's recommendation, intended to filter smaller, non-human heat signatures. We compare the person-present agreement rate before and after the change using the same coding protocol throughout, with no other change to the extension logic or the observation procedure.

Results

Across the 96-booking sample, the floodlight controller logged 132 extension events. Cross-referencing each event's onset against the supervisor's independent closing-log entry, a person was confirmed still on the ground for 61 events (46%). The remaining 71 events (54%), adjudicated from the maintenance camera's stills, split across grounds-maintenance activity (25 events, 19%, chiefly a late mowing or line-marking pass finishing after the booking's advertised end), an irrigation-cycle start (21 events, 16%), wildlife at the oval's verge (15 events, 11%, kangaroos grazing the verge and one resident fox accounting for most sightings), and an insect swarm visible around a lamp housing (10 events, 8%).

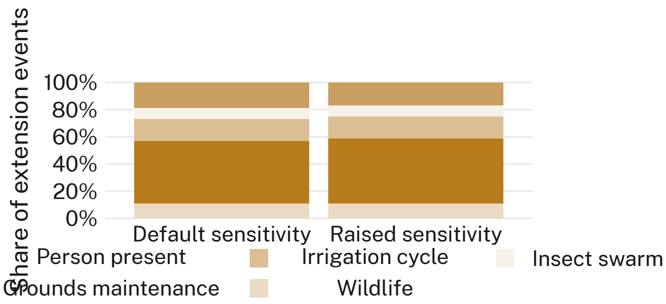

Figure 1: Composition of extension-event causes is materially unchanged by the sensitivity-threshold ablation: 46% person-confirmed before, 48% after.

Raising the sensitivity threshold after week 7 moved the overall agreement rate from 46% to 48%, a two-percentage-point shift we treat as not meaningfully different from no change at all (χ^2 across the five-category breakdown, $p > 0.3$). The shift in non-human causes was similarly marginal — grounds-maintenance's share fell from 19% to 17% and wildlife's held at 11% — consistent with the threshold change filtering a small, roughly cause-neutral slice of triggers rather than sharpening the sensor's read on any one of them. The pattern holds across the working week: no weekday's agreement rate moved by more than three points after the change (Figure 2).

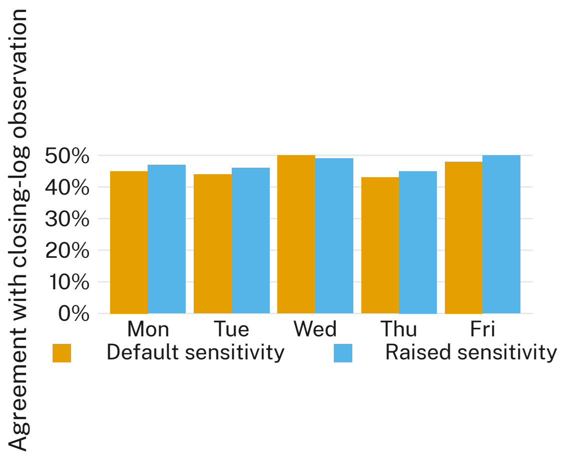

Figure 2: Agreement rate by weekday and sensitivity setting: the ablation changes nothing on any given night either.

The weekly agreement rate ranged from 38% to 54% across the window with no visible trend across the teaching period and no discontinuity at week 8, when the threshold changed (Figure 3). A campus open day in week 4, which brought unrelated evening foot traffic within sight of two of the four towers, coincided with the window's single highest weekly rate (52%) but not with a corresponding rise in confirmed-occupant events specifically — the open-day evening's extra extensions were, on inspection, disproportionately a grounds-crew clean-up pass ahead of the event rather than lingering attendees.

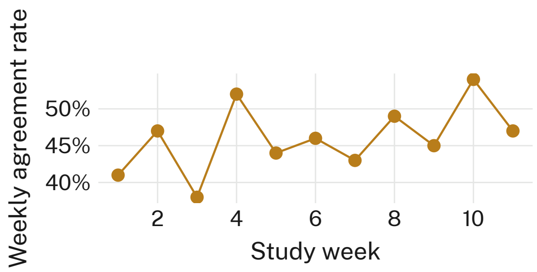

Figure 3: Weekly agreement rate, weeks 1–11. The threshold change after week 7 leaves no visible mark on the series.

Coded cause	Events	Share
Person present	61	46%
Grounds maintenance	25	19%
Irrigation cycle	21	16%
Wildlife	15	11%
Insect swarm	10	8%

Table 1: Coding scheme applied to all 132 extension events, pooled across both sensitivity settings.

Discussion

The extension logic does what it was built to do: it almost never leaves a player in the dark. It does this, our data suggest, largely by staying on for reasons that have nothing to do with a player. Fewer than half of extension events coincide with a confirmed occupant, and the remainder trace to a small, stable set of non-human triggers — a groundskeeping pass, a sprinkler cycle, the verge's kangaroos, the insects a floodlight tower reliably attracts — that the sensor array has no way to tell apart from a lingering team. Raising the sensitivity threshold, the fix the installer already recommends for nuisance triggers, does not change this: our results suggest the gap sits upstream of any single threshold, in what the sensor array is asked to represent rather than how finely it is tuned. A feature that reliably avoids the complaint it was built for while answering a different, unaudited question is not obviously a fault; whether it should be treated as one likely depends on what the extension is judged to cost, and no one currently asks the closing log that question.

Limitations

Our sample covers a single oval across one eleven-week teaching period, and the coding scheme's five categories necessarily compress a messier set of possible causes into convenient buckets; a sixth category, or a wetter winter, might redistribute the 54% differently, though the consistency of that share across eleven weeks and two sensitivity settings suggests the redistribution would be modest. The supervisor's closing log is a manually completed record rather than an automated one, though its independence from the floodlight controller's own sensing loop is exactly the property our design required, and inter-coder agreement on the camera stills was high (89%) wherever the two records disagreed. The sensitivity-threshold ablation was a single, contractor-scheduled change rather than a randomised manipulation, though its timing was fixed well before we saw the pre-change data, which limits any suggestion that we shaped the comparison to fit.

Conclusion

A motion-sensing floodlight extension answers exactly the question it was built to answer — is something moving under the towers — and that question turns out to be a looser proxy for "is a player still here" than the feature's name implies. Pairing the controller's own log against an independent closing-time observation is a cheap addition to a facility that, like this one, already keeps a paper closing log for other reasons; we suspect a similar pairing would surface a similar gap wherever a courtesy extension shares its detection zone with grounds equipment, irrigation, or wildlife. Future work might extend the design to the University's other lit outdoor recreation spaces, and ask whether the gap narrows there or simply relocates.

References

[1] A. Shokrollahi, J. A. Persson, R. Malekian, A. Sarkheyli-Hägele, and F. Karlsson, "Passive Infrared Sensor-Based Occupancy Monitoring in Smart Buildings: A Review of Methodologies and Machine Learning Ap-

proaches," *Sensors*, 2024, doi: 10.3390/s24051533.

[2] P. Chaudhari, Y. Xiao, M. M.-C. Cheng, and T. Li, "Fundamentals, Algorithms, and Technologies of Occupancy Detection for Smart Buildings Using IoT Sensors," *Sensors*, 2024, doi: 10.3390/s24072123.

[3] T. Futagami and N. Hayasaka, "Experimental evaluation of occupancy lighting control based on low-power image-based motion sensor," *SICE Journal of Control, Measurement, and System Integration*, 2021, doi: 10.1080/18824889.2021.1987635.

[4] T. Futagami, "Experimental evaluation for occupancy lighting control using combination of PIR and image-based sensors," *SICE Journal of Control, Measurement, and System Integration*, 2024, doi: 10.1080/18824889.2024.2310870.

[5] Z. Pang *et al.*, "Long-Term field testing of the accuracy and HVAC energy savings potential of occupancy presence sensors in a single-family home," *Energy and Buildings*, 2025, doi: 10.1016/j.enbuild.2024.115161.

[6] B. Kula, D. Mitra, Y. Chu, K. Cetin, R. Gallagher, and S. Banerji, "Laboratory testing methods to evaluate the reliability of occupancy sensors for commercial building applications," *Building and Environment*, 2023, doi: 10.1016/j.buildenv.2023.110457.

[7] Y. Chu, D. Mitra, K. Cetin, N. Lajnef, F. Altay, and S. Velipasalar, "Development and testing of a performance evaluation methodology to assess the reliability of occupancy sensor systems in residential buildings," *Energy and Buildings*, 2022, doi: 10.1016/j.enbuild.2022.112148.

[8] M. Mahoor, Z. S. Hosseini, A. Khodaei, A. Paaso, and D. Kushner, "State-of-the-art in smart streetlight systems: a review," *IET Smart Cities*, 2020, doi: 10.1049/iet-smc.2019.0029.

[9] A. S. Glen, S. Cockburn, M. Nichols, J. Ekanayake, and B. Warburton, "Optimising Camera Traps for Monitoring Small Mammals," *PLOS ONE*, 2013, doi: 10.1371/journal.pone.0067940.

[10] F. Cunha, E. M. dos Santos, R. Barreto, and J. G. Colonna, "Filtering Empty Camera Trap Images in Embedded Systems," *arXiv preprint arXiv:2104.08859*, 2021.

[11] M. Sefawe, S. Ganti, J. Segalla, E. He, I. Tourner, and J. Gersey, "A Structured Review of Fixed and Multimodal Sensing Techniques for Bat Monitoring," *arXiv preprint arXiv:2512.07153*, 2025.

[12] A. C. S. Owens and S. M. Lewis, "The impact of artificial light at night on nocturnal insects: A review and synthesis," *Ecology and Evolution*, 2018, doi: 10.1002/ece3.4557.

[13] J. Bennie, T. W. Davies, D. Cruse, R. Inger, and K. J. Gaston, "Cascading effects of artificial light at night: resource-mediated control of herbivores in a grassland ecosystem," *Philosophical Transactions of the Royal Society B: Biological Sciences*, 2015, doi: 10.1098/rstb.2014.0131.

[14] J. M. Katabaro, Y. Yan, T. Hu, Q. Yu, and X. Cheng, "A review of the effects of artificial light at night in urban areas on the ecosystem level and the remedial measures," *Frontiers in Public Health*, 2022, doi: 10.3389/fpubh.2022.969945.

[15] I. Qaisar, K. Sun, Q. Jia, and Q. Zhao, "Experimental study on surveillance video-based indoor occupancy measurement with occupant-centric control," *arXiv preprint arXiv:2603.26081*, 2026.

[16] O. Vandermeer and C. Beng, "Waved and Believed: Boost Wiring, Indicator-Light Feedback, and Hand-Drying Dwell Time at a Campus Hand Dryer," *Slop University technical report*, 2026, doi: 10.5555/slop.3mnp9h.

[17] J. Nwosu and S. Okwuosa, "Swapped the Card, Not the Habit: Session Length and Peak Occupancy in the Gym Sauna Before and After a Safety-Scare Timer Swap," *Slop University technical report*, 2026, doi: 10.5555/slop.8qz26v.